

Elimination of Adversarial Strategies from Population Game Dynamics via Partial Feedback Linearization Penalty Control

Jack Pearson[†]

Brendon G. Anderson[†]

Abstract—We propose a new framework for modeling, analyzing, and eliminating adversarial behaviors, such as cheating, in population games and evolutionary dynamics. Our model accounts for the probability of a cheater being caught, as well as the penalty incurred by a cheater when they are caught. We provide a complete theoretical analysis of the various possible Nash equilibria of such games with cheating, including necessary and sufficient conditions for their existence. We leverage our analyses to design a partial feedback linearization control scheme, utilizing the cheating penalty as the control variable, which we prove is capable of eliminating the usage of the adversarial strategy when the game plays out according to the infamous replicator dynamics. We prove that our control scheme naturally inherits near-optimality, in the sense that the penalty stabilizes to a merciful and judicious value at nearly the minimum penalty necessary to grant no cheating at equilibrium, and we explicitly identify the suboptimality gap. Simulations of the classical Rock-Paper-Scissors game, augmented with a cheating strategy, are conducted to illustrate and corroborate our theory.

I. INTRODUCTION

In this paper, we consider evolutionary games in which a large population of players strategically interact with one another and continually revise their choice of strategy [1, 2]. The strategy revision is performed according to some “revision protocol,” aimed at maximizing expected payoff, and this leads to the population’s distribution of strategy choices to evolve according to some dynamics. One of the most well-known examples of such evolutionary dynamics is the renowned “replicator dynamics” model [3]. Such models are widely used across engineering and science applications, including traffic networks, economic markets, biological systems, and even nonconvex optimization [1, 4].

Extensive related work has analyzed both cooperation, and selfish or adversarial behaviors in social dilemmas [5–9], with public goods games, the prisoner’s dilemma, and Braess’ paradox offering famous examples. In this paper, we are primarily concerned with adversarial behaviors that act in the same manner as “cheating” does [10]. Some prior work has studied the evolutionary interactions between cooperators and cheaters, most notably in a biological settings [11–14]. Related work has also considered control-theoretic mechanisms for

promoting cooperation [15–17]. In particular, the work [18] develops an optimal incentive mechanism, which combines rewards for cooperators and punishments for defectors, to steer a population toward full cooperation. These cooperative models, however, define undesirable or adversarial behavior within the base structure of the particular social dilemma at hand. As a result, they do not provide an adequate framework for augmenting the model with external behaviors outside of the basic rules of the game, which is necessary when considering adversarial strategies such as cheating.

In this paper, we propose a framework to augment population game models with external adversarial behaviors, which we refer to as cheating, that are outside of the rules of the base game. One key novelty of our framework, compared to standard population games and the aforementioned related works, is the incorporation of a model for the cheaters getting caught with some probability, as well as the payoffs to cheaters depending on whether they get caught or not. Leveraging our framework, we develop a control-theoretic disincentive mechanism to steer the population away from using adversarial strategies. Our main contributions are:

- 1) We develop a framework for augmenting linear population games with an external adversarial strategy (i.e., cheating).
- 2) We characterize the Nash equilibria of the augmented game, providing necessary and sufficient conditions for the existence of pure cheating, no cheating, and partial cheating equilibria. We construct examples that demonstrate the possible coexistence of all three types of equilibria.
- 3) We propose a control scheme that assigns penalties to cheating which asymptotically eliminates adversarial behavior and recovers the base game’s dynamics. We prove that the controlled penalties stabilize near the minimum necessary value, indicating a near-optimality interpreted as “judicious” punishment that is not unnecessarily harsh.
- 4) We illustrate and corroborate our theory with numerical simulations of the classical Rock-Paper-Scissors game, augmented with cheating.

To streamline exposition, all proofs are deferred to the appendix.

[†]Mechanical Engineering Department, California Polytechnic State University. Emails: {jpearson11, bga}@calpoly.edu.

II. PROPOSED MODEL OF ADVERSARIAL GAMEPLAY

We consider the population game setting in which a large-scale society of players partake in a game with $n \in \mathbb{N}$ basic strategies, $s_1, \dots, s_n$. We follow the conventions of population game theory [1] and model the society as a homogeneous continuum of players, allowing the state of the society to be modeled by the aggregate proportion of players employing each strategy. Specifically, a population state of the n -strategy base game is given by a vector

$$\tilde{x} \in \Delta^{n-1} := \{\tilde{y} \in \mathbb{R}^n : \tilde{y} \geq 0, \sum_{i=1}^n \tilde{y}_i = 1\},$$

with $\tilde{x}_i \in [0, 1]$ representing the proportion of players choosing strategy s_i , and Δ^{n-1} being the $(n-1)$ -dimensional probability simplex. The base game is a function $\tilde{F}: \Delta^{n-1} \rightarrow \mathbb{R}^n$, defined such that $\tilde{F}_i(\tilde{x}) \in \mathbb{R}$ encodes the expected payoff to a player who chooses to play strategy s_i when the society is at the population state $\tilde{x}$. Throughout this work, we assume that the base game's payoff structure is linear, which is common in practice: $\tilde{F}(x) = \tilde{A}\tilde{x}$, for some $\tilde{A} \in \mathbb{R}^{n \times n}$.

To model adversarial gameplay, which we also refer to as “cheating” throughout, we augment the base game with $m \in \mathbb{N}$ additional strategies, $s_{n+1}, \dots, s_{n+m}$, each of which corresponds to some way of cheating. The overall strategy set therefore becomes $S = \{s_1, \dots, s_n, s_{n+1}, \dots, s_{n+m}\}$. The population state of the augmented game, with cheating included, expands to a higher-dimensional probability vector $x \in \Delta^{n+m-1}$. The augmented game is now a function $F: \Delta^{n+m-1} \rightarrow \mathbb{R}^{n+m}$ with $F_i(x)$ representing the expected payoff to strategy s_i (possibly a cheating strategy) when the society is at the population state x . For simplicity going forward, we assume that $m = 1$, meaning there is a single cheating strategy, s_{n+1} .

The novelty that distinguishes our model from a conventional population game, one which simply treats the cheating strategy as a base strategy of a larger population game, is given by the following additional structure that we impose onto our model:

Assumption 1. There exists a probability $p \in [0, 1]$, corresponding to the probability that a player gets caught when employing the cheating strategy s_{n+1} , such that the augmented game takes the following form:

$$F(x) = Ax, \quad A = \begin{bmatrix} \tilde{A} & \alpha \mathbf{1}_n \\ \beta \mathbf{1}_n^\top & \gamma \end{bmatrix}, \quad (1)$$

where

$$\begin{aligned} \alpha &:= pw - (1-p)l, \\ \beta &:= -pc + (1-p)w, \\ \gamma &:= -p(pd_c + (1-p)c) + (1-p)(pw + (1-p)d_w), \end{aligned}$$

for some positive $w, l, c \in \mathbb{R}$ and some nonnegative $d_w, d_c \in \mathbb{R}$, with $\mathbf{1}_n$ denoting the n -vector of all ones.

The augmented game structure imposed in Assumption 1 is interpreted as follows. Expected payoffs $A_{ij} = \tilde{A}_{ij}$ to strategies $i \in \{1, \dots, n\}$ abiding by the base game's rules, and played against those $j \in \{1, \dots, n\}$ that also abide by the rules, are governed by the base game's payoff matrix $\tilde{A}$. Payoffs to strategies $i \in \{1, \dots, n\}$ abiding by the base game's rules, and played against cheating s_{n+1} , will win a value w with probability p (when the cheater is caught) and lose a value $-l$ with probability $1-p$ (when the cheater is not caught), resulting in the expected payoff $A_{i,n+1} = \alpha = pw - (1-p)l$. Similarly, payoffs to cheating s_{n+1} , played against strategies $i \in \{1, \dots, n\}$ abiding by the base game's rules, will lose a value c with probability p (when the cheater is caught) and win a value w with probability $1-p$ (when the cheater is not caught), resulting in the expected payoff $A_{n+1,i} = \beta = -pc + (1-p)w$. Finally, d_c and d_w correspond to the draw values granted to a cheater playing against a cheater when both are caught and both are not caught, respectively, resulting in the expected payoff value $A_{n+1,n+1} = \gamma$ in (1). In many practical examples, one has $c > l$ to reflect a harsh penalty of getting caught cheating. Similarly, often $d_w = 0$ to reflect two cheaters acting as if they follow usual draw rules (i.e., zero payoff) when neither of them is caught, and $d_c = 0$ to reflect two mutually caught cheaters keeping their cheating secret from others.

The cheating strategy, when not caught, will always win the competition with a value of w against any of the pure, non-cheating strategies. In this sense, cheating functions similarly to the classical game-theoretic notion of a dominant strategy. Dominant strategies have been extensively studied in the literature, with some works explicitly considering variants of the base game augmented with dominant strategies [19]. However, if cheating were modeled as a dominant strategy, every player would cheat since cheating would always have the best payoff. Clearly, this is not a realistic model of cheating. In real-world circumstances, cheaters can be caught and incur (often large) undesirable losses. Dominant strategy models do not come equipped with this notion of “getting caught,” and are therefore inadequate for our purposes, motivating the need for our proposed model of cheating in which the expected payouts depend on the probability of cheaters getting caught.

A. Replicator Dynamics

In this work, we will consider evolutionary models to account for iterative gameplay where players continually revise their choice of strategy over time according to some “revision protocol” [1]. We choose to study the augmented game's evolution, with a particular focus on

the usage of adversarial gameplay, under the framework of the “replicator dynamics” [3], which is one of the most well-known and studied models in evolutionary game theory. The replicator dynamics are given by the following system of $n+1$ ordinary differential equations:

$$\dot{x}_i(t) = x_i(t) (F_i(x(t)) - x(t)^\top F(x(t))), \quad (2)$$

for $i \in \{1, \dots, n+1\}$. In Section IV, we will design a feedback control law that, when coupled with the replicator dynamics of the augmented game, succeeds in eliminating cheating in the sense that $x_{n+1}(t) \xrightarrow{t \rightarrow \infty} 0$.

B. Running Example: Rock-Paper-Scissors-Cheat

Throughout the remainder of this work, we will use the infamous Rock-Paper-Scissors (RPS) game as a representative example to illustrate our framework and results. We choose to apply our methods to RPS due to the game’s extensive prior analyses and its intuitive interpretations.

The basic 3-strategy RPS game with $s_1 = \text{Rock}$, $s_2 = \text{Paper}$, and $s_3 = \text{Scissors}$ is defined by the payoff matrix

$$\tilde{A} = \begin{bmatrix} 0 & -l & w \\ w & 0 & -l \\ -l & w & 0 \end{bmatrix}.$$

Here, $w > 0$ and $l > 0$ are the values associated with winning and losing an instance of Rock-Paper-Scissors, respectively, which we assume are identical to the win and loss values w and l associated with cheating (without being caught) found in the augmented payoff matrix (1). By augmenting RPS with an adversarial strategy according to our model (1), we arrive at the 4-strategy game Rock-Paper-Scissors-Cheat (RPSC). The cheating strategy can be interpreted as follows: in basic RPS, the players must close their eyes until both players are ready to unveil their choice of Rock, Paper, or Scissors. In RPSC, cheating amounts to “peeking” at the opponent’s choice, and subsequently choosing to play whichever strategy ensures the win.

III. NASH EQUILIBRIUM ANALYSIS OF THE AUGMENTED GAME

In this section, we analyze the Nash equilibria of games augmented with an adversarial strategy. We identify and characterize the Nash equilibria into three categories, namely, 1) pure cheating, 2) no cheating, and 3) partial cheating, and we derive necessary and sufficient conditions for each type of equilibrium’s existence.

Definition 1. Consider a generic n -strategy population game $F: \Delta^{n-1} \rightarrow \mathbb{R}^n$. A population state $x \in \Delta^{n-1}$ is a Nash equilibrium of F if

$$F_i(x) = \max_{j \in \{1, \dots, n\}} F_j(x)$$

for all $i \in \{1, \dots, n\}$ such that $x_i > 0$. The set of all Nash equilibria of F is denoted by $\text{NE}(F)$.

It is straightforward to show that a state $x \in \Delta^{n-1}$ is a Nash equilibrium of a population game F if and only if $y^\top F(x) \leq x^\top F(x)$ for all $y \in \Delta^{n-1}$.

Example 1. The unique Nash equilibrium of the basic RPS game is $\tilde{x} = (1/3, 1/3, 1/3)$, reflecting the symmetry in the Rock-Paper-Scissors payoff structure [1, Example 3.3.2].

Going forward, we reserve the notations $\tilde{F}$ and F to refer to the base game and augmented game of interest, respectively. We also use e_i to denote the i th standard unit vector, with dimension inferred from context.

A. Characterizing the Possible Nash Equilibria

We begin our analysis by fully characterizing the set of possible Nash equilibria of the augmented game.

Theorem 1. If $x \in \text{NE}(F)$, then x must lie on the line segment between the pure cheating population state e_{n+1} and some Nash equilibrium of the base game $\tilde{F}$:

$$x = (1 - \theta)(\tilde{x}, 0) + \theta e_{n+1}, \quad (3)$$

for some $\theta \in [0, 1]$ and some $\tilde{x} \in \text{NE}(\tilde{F})$.

Theorem 1 shows that all Nash equilibria of the game with cheating must exist within the cone generated by the pure cheating population state e_{n+1} and the set $\text{NE}(\tilde{F})$ of Nash equilibria of the base game, lifted into $\Delta^n \subseteq \mathbb{R}^{n+1}$. This characterizes the set of possible Nash equilibria of F into three cases, illustrated in Figure 1: 1) Pure cheating (colored red), 2) Nash equilibrium gameplay of $\tilde{F}$ without any cheating (colored blue), and 3) a (convex) mixture of cheating with Nash gameplay of $\tilde{F}$ (colored orange). The partial cheating Nash equilibria of F can be interpreted as follows: at Nash equilibrium, the subset of the population that chooses not to cheat will play according to the base game’s Nash equilibrium, amongst themselves. This implies that the augmentation of the game with an adversarial strategy *cannot* influence how non-cheaters play amongst themselves at equilibrium, only whether or not they will abandon the baseline rules of the game in favor of cheating.

Now that we know, geometrically, where all viable Nash equilibria of the augmented game can be possibly located, we derive necessary and sufficient conditions for each of the three types of Nash equilibria to exist.

B. Pure Cheating Nash Equilibrium

Below, we show that the state e_{n+1} , corresponding to the entire population cheating, is a Nash equilibrium

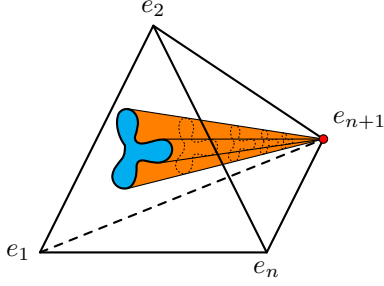


Fig. 1: Illustration of the set $\text{NE}(F)$ of all possible Nash equilibria of the augmented game F with cheating. The pure cheating population state e_{n+1} is shown in red. The set $\text{NE}(\tilde{F}) \times \{0\}$ of Nash equilibria of the base game $\tilde{F}$ without cheating is shown in blue. The orange set illustrates mixtures of Nash equilibria of the base game with some partial amount of cheating.

when the penalty associated with getting caught cheating, or the probability of getting caught cheating, is sufficiently small. Thus, cheating can introduce a monomorphic (vertex) Nash equilibrium when augmented to basic games that have no monomorphic Nash equilibria, such as RPS. We start by characterizing the existence of the pure cheating Nash equilibrium in terms of the cost c associated with getting caught cheating:

Theorem 2. *Consider the pure cheating population state $x = e_{n+1}$. Assume that $p \in (0, 1)$. It holds that x is a Nash equilibrium of the game F if and only if*

$$c \leq \frac{(1-p)l + (1-p)^2 d_w - (w + d_c)p^2}{p(1-p)}. \quad (4)$$

Theorem 2 shows that the pure cheating Nash equilibrium is eliminated whenever the penalty c is set to be larger than an appropriate threshold. If l is small enough, then the threshold on the right of (4) can become nonpositive and hence inexistence of the pure cheating Nash equilibrium trivially holds no matter the value of $c > 0$. Intuitively, this reflects players' motivation to play the game according to the rules whenever the risks associated with genuinely losing are small enough. Interestingly, even though the win payoff w appears positively in the average payoffs to both cheaters (i.e., the entries $A_{n+1,j}$ for all j) and noncheaters (i.e., the entries $A_{i,n+1}$ for all i) alike, we see that inexistence of the pure cheating Nash equilibrium also trivially holds whenever w is sufficiently large. Therefore, a large value w motivates players to achieve that payoff through genuine gameplay, rather than through cheating.

We remark that our results in this section are independent of the base game payoff matrix $\tilde{A}$, and therefore we conclude that the existence of the pure cheating Nash equilibrium is *independent of the rules of the base game*.

C. No Cheating Nash Equilibrium

We now consider the Nash equilibrium state where no cheating exists. Unlike the existence of a pure cheating Nash equilibrium, we find below that the existence of a no cheating Nash equilibrium inherently depends on the base game's payoff values.

Theorem 3. *Let $x = (\tilde{x}, 0)$ be a population state without cheating, and with $\tilde{x} \in \text{NE}(\tilde{F})$. It holds that x is a Nash equilibrium of F if and only if*

$$-pc + (1-p)w \leq \tilde{x}^\top \tilde{F}(\tilde{x}). \quad (5)$$

Intuitively, Theorem 3 states that there will be no cheating at Nash equilibrium when the expected payoff to genuine gameplay (according to the base game's rules) is larger than the expected payoff to a cheater playing against someone who plays by the rules. The inequality (5) shows that cheating can be eliminated from equilibrium by increasing either p or c , or by decreasing the value w won by a cheater who is not caught. This motivates the usage of these variables as control inputs to steer the population towards an equilibrium state without cheating; we design such a control law for the cost c in Section IV. Contrarily, the genuine loss value l against a cheater and the draw values d_w and d_c do not influence the existence of a no cheating Nash equilibrium.

Example 2. Applying Theorem 3 to RPSC, we find that the symmetric no cheating population state $x = (1/3, 1/3, 1/3, 0)$ is a Nash equilibrium of the augmented game if and only if

$$c \geq \frac{-3wp + (2w + l)}{3p}. \quad (6)$$

Interestingly, this trivially holds true, even for an arbitrarily small cheating penalty $c > 0$, whenever

$$p \geq \frac{2w + l}{3w}.$$

This shows that with a sufficient amount of surveillance, cheating need not even be penalized at all; the population enjoys a no cheating equilibrium through what can be thought of as natural self-regulation. In other words, a high degree of supervision, a sufficiently large genuine win value w , and a sufficiently small genuine loss value l , together, provide enough incentive for players to not cheat. Unlike the penalty c which has the possibility to be arbitrarily small and still result in a no cheating equilibrium, rearranging (6) shows that one needs $p \geq \frac{2w+l}{3(w+c)} > 0$ in order to attain a no cheating equilibrium; given fixed win and loss values w and l , there is no way to forgo surveillance entirely unless $c \rightarrow \infty$.

D. Partial Cheating Nash Equilibrium

We now provide conditions under which Nash equilibria of the augmented game exist in the relative interior of Δ^n , meaning that some of the population, but not all, chooses to cheat.

Theorem 4. *If x is a Nash equilibrium of F in the relative interior of Δ^n , then it must take the form (3) for some $\tilde{x} \in \text{NE}(\tilde{F})$, with $\theta \in (0, 1)$ satisfying the affine equation*

$$\mathcal{A}\theta = b. \quad (7)$$

where

$$\begin{aligned} \mathcal{A} &:= p^2 c + \left(\frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x} - (1 - p + p^2)w \right. \\ &\quad \left. + (1 - p)l + (1 - p)^2 d_w - p^2 d_c \right), \\ b &:= pc + \left(\frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x} - (1 - p)w \right). \end{aligned}$$

In this case, either the relative interior Nash equilibrium is unique with

$$\theta = \mathcal{A}/b \in (0, 1), \quad (8)$$

or every point on the line segment $\{(1 - \theta)(\tilde{x}, 0) + \theta e_{n+1} : \theta \in (0, 1)\}$ is a Nash equilibrium of F .

On the other hand, if $\tilde{x} \in \text{NE}(\tilde{F})$ and if $\theta \in (0, 1)$ satisfies (7), then x given by (3) is a relative interior Nash equilibrium of F .

Theorem 4 shows that three possibilities may occur when it comes to the existence of partial cheating Nash equilibria corresponding to some particular base Nash gameplay $\tilde{x}$: 1) there may be no partial cheating equilibria, 2) there may be a unique partial cheating equilibrium, or 3) there may be a continuous line segment of partial cheating equilibria.

Given a reference Nash equilibrium $\tilde{x}$ of the base game $\tilde{F}$, the following corollary gives a computationally tractable method for verifying that $\tilde{x}$ is not susceptible to having some amount of cheaters mixed into the population when at equilibrium of the augmented game.

Corollary 1. *Consider a Nash equilibrium $\tilde{x} \in \text{NE}(\tilde{F})$ of the base game $\tilde{F}$. If θ defined by (8) is a well-defined real number and $\theta \notin (0, 1)$, then there are no Nash equilibria of the augmented game F belonging to the strict line segment between $(\tilde{x}, 0)$ and e_{n+1} .*

Should θ in (8) be calculated and shown to be outside of $(0, 1)$ for every base game Nash equilibrium $\tilde{x} \in \text{NE}(\tilde{F})$, then Corollary 1 ensures that the only possible Nash equilibria of F are pure cheating e_{n+1} and the base game Nash equilibria $(\tilde{x}, 0)$ without any cheating.

Example 3. Consider the RPSC game and assume that $p \in (0, 1)$. A straightforward calculation shows that θ ,

as defined in (8), satisfies $0 < \theta < 1$ if and only if $c_1 < c < c_2$, where

$$\begin{aligned} c_1 &:= \frac{-3pw + (2w + l)}{3p} \\ c_2 &:= \frac{(1 - p)l + (1 - p)^2 d_w - (w + d_c)p^2}{p(1 - p)}. \end{aligned}$$

These lower and upper bounds on the penalty c exactly coincide with the thresholds at which no cheating and pure cheating states become Nash equilibria, as granted by (6) and Theorem 2, respectively. Supplementing this finding with the conclusions of Theorem 4, we conclude that there are five possible configurations for the Nash equilibria of RPSC:

- 1) If $c < c_1$, then e_{n+1} is the only Nash equilibrium.
- 2) If $c_1 < c < c_2$, then there are exactly three Nash equilibria: e_{n+1} , $(1/3, 1/3, 1/3, 0)$, and the relative interior Nash equilibrium identified in Theorem 4.
- 3) If $c > c_2$, then $(1/3, 1/3, 1/3, 0)$ is the only Nash equilibrium.
- 4) If $c = c_1 = c_2$, then $\text{NE}(F)$ is the entire line segment between e_{n+1} and $(1/3, 1/3, 1/3, 0)$, including the endpoints.
- 5) If $c = c_1 < c_2$ or $c = c_2 > c_1$, then there are exactly two Nash equilibria: e_{n+1} and $(1/3, 1/3, 1/3, 0)$.

Thus, we find that, even though the RPS game had a unique Nash equilibrium, the introduction of a cheating strategy can result in the augmented game simultaneously having multiple Nash equilibria which can take on a disconnected, nonconvex geometry.

Example 4. We give an example of the fourth possible configuration for RPSC Nash equilibria identified in Example 3. Let $w = l = c > 0$, $p = 1/2$, and $d_w = d_c = 0$. The payoff matrix reduces to

$$A = \begin{bmatrix} 0 & -c & c & 0 \\ c & 0 & -c & 0 \\ -c & c & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

We observe that

$$\begin{aligned} F_i(1/3, 1/3, 1/3, 0) &= F_i(e_4) \\ &= F_i(\theta(1/3, 1/3, 1/3, 0) + (1 - \theta)e_4) = 0 \end{aligned}$$

for all $\theta \in [0, 1]$ and all strategies $i \in \{1, \dots, 4\}$, cheating and non-cheating alike. Therefore, every point on the line segment between the no cheating state $(1/3, 1/3, 1/3, 0)$ and the pure cheating state e_4 , including these endpoints, is a Nash equilibrium. This is due to the symmetry of the parameters w, l, c, p, d_w, d_c , which can be intuitively thought of as making players indifferent to cheating or playing standard RPS.

Example 5. We give an example of the fourth possible configuration for RPSC Nash equilibria identified in Example 3. Let $w = 2$, $l = 1$, $c = 2$, $p = 1/2$, $d_w = 3$, and $d_c = 0$. The payoff matrix reduces to

$$A = \begin{bmatrix} 0 & -1 & 2 & 1/2 \\ 2 & 0 & -1 & 1/2 \\ -1 & 2 & 0 & 1/2 \\ 0 & 0 & 0 & 3/4 \end{bmatrix}.$$

It is a straightforward computation to verify that $(1/3, 1/3, 1/3, 0)$, $(0, 0, 0, 1)$, and

$$\left(\frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{4}{7}\right) = \frac{3}{7} \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0\right) + \frac{4}{7}(0, 0, 0, 1)$$

are all Nash equilibrium. Yet, for any other state $x = (1 - \theta)(1/3, 1/3, 1/3, 0) + \theta(0, 0, 0, 1)$ with $\theta \notin \{0, 4/7, 1\}$, we see that $F_1(x) = \frac{1}{3} + \frac{1}{6}\theta \neq \frac{3}{4}\theta = F_4(x)$, so such states are not Nash equilibria.

IV. FEEDBACK LINEARIZATION CONTROL OF ADVERSARIAL PLAY

In this section, we consider the cheating penalty c to be a time-varying control variable. We posit that this is a realistic choice of control variable, as agents involved in regulating the usage of adversarial behaviors in societal-scale games often have authority over the penalties sentenced to those that break the rules. From this perspective, the goal of this section is to develop a control law that automatically computes appropriate penalties c that will steer the population towards a no cheating equilibrium as the game unfolds according to the replicator dynamics, i.e., the control law should eliminate adversarial gameplay.

Although the replicator dynamics are nonlinear, they are control-affine in the control variable c :

$$\begin{aligned} \dot{x}_i(t) &= x_i(t)(g_i^\top x(t) - x(t)^\top Gx(t)) \\ &\quad + x_i(t)(h_i^\top x(t) - x(t)^\top Hx(t))c(t), \end{aligned}$$

for $i \in \{1, \dots, n+1\}$, where G and H are the unique matrices such that $A = G + cH$, i.e.,

$$\begin{aligned} G &= \begin{bmatrix} \tilde{A} & (pw - (1-p)l)\mathbf{1}_n \\ (1-p)w\mathbf{1}_n^\top & -p^2d_c + (1-p)(pw + (1-p)d_w) \end{bmatrix}, \\ H &= \begin{bmatrix} \mathbf{0}_{n \times n} & \mathbf{0}_n \\ -p\mathbf{1}_n^\top & -p(1-p) \end{bmatrix}, \end{aligned}$$

and where $g_i \in \mathbb{R}^{n+1}$ and $h_i \in \mathbb{R}^{n+1}$ respectively denote the i th rows of G and H , and where $\mathbf{0}_{n \times n}$ denotes the $n \times n$ matrix of all zeros.

We propose to introduce an auxiliary control variable u and to employ the following feedback control law

$$c(t) = \frac{u(t) - x_{n+1}(t)(g_{n+1}^\top x(t) - x(t)^\top Gx(t))}{x_{n+1}(t)(h_{n+1}^\top x(t) - x(t)^\top Hx(t))}, \quad (9)$$

to linearize the dynamics of the cheating strategy, i.e.,

$$\dot{x}_{n+1}(t) = u(t)$$

in closed-loop. We then propose to use the partial state feedback controller

$$u(t) = -kx_{n+1}(t), \quad (10)$$

where the control gain $k > 0$ is constant. When employing our feedback control law, we do not explicitly constrain $c(t)$ to be positive, although we find in practice that $c(t)$ always remains positive in our simulations. Clearly, this partial feedback linearization strategy succeeds in eliminating adversarial gameplay, since $x_{n+1}(t) = x_{n+1}(0)e^{-kt} \xrightarrow{t \rightarrow \infty} 0$. However, it is not immediately clear how the other strategies evolve. We now show that they fall back into the base game's dynamics as if cheating was never an option at all.

Theorem 5. Let $\tilde{f}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the vector field defining the replicator dynamics of the base game, and let $f: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ be the vector field defining the closed-loop replicator dynamics of the augmented game with the control law (9) and (10) with $k > 0$:

$$\begin{aligned} \tilde{f}_i(\tilde{x}) &= \tilde{x}_i \left(\tilde{F}_i(\tilde{x}) - \tilde{x}^\top \tilde{F}(\tilde{x}) \right), \\ f_j(x) &= x_j \left(g_j^\top x - x^\top Gx \right) \\ &\quad - x_j \left(h_j^\top x - x^\top Hx \right) \frac{g_{n+1}^\top x - x^\top Gx + k}{h_{n+1}^\top x - x^\top Hx}, \end{aligned}$$

for $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, n+1\}$. Define the shorthand notation $x_{1:n} = (x_1, \dots, x_n)$. If $p > 0$, then the closed-loop trajectory of the augmented replicator dynamics satisfies both of the following:

- 1) $\lim_{t \rightarrow \infty} |c(t) - \tilde{c}(t)| = 0$, where

$$\tilde{c}(t) := \frac{(1-p)w - x_{1:n}(t)^\top \tilde{A}x_{1:n}(t) + k}{p}, \quad (11)$$

and

- 2) $\lim_{t \rightarrow \infty} |f_i(x(t)) - \tilde{f}_i(x_1(t), \dots, x_n(t))| = 0$ for all $i \in \{1, \dots, n\}$.

Corollary 2. Suppose that $\text{NE}(\tilde{F})$ is globally attracting under the base game's replicator dynamics. Then, the closed-loop replicator dynamics of the augmented game with the control law (9) and (10) converge to a lifted Nash equilibrium $(\tilde{x}, 0)$, $\tilde{x} \in \text{NE}(F)$, with no cheating:

$$\lim_{t \rightarrow \infty} x(t) = (\tilde{x}, 0).$$

Furthermore, the time-varying penalty assigned to cheating converges to the minimum value necessary to ensure the no cheating equilibrium, determined by (5) in Theorem 3, offset by a constant value of k/p :

$$\lim_{t \rightarrow \infty} c(t) = \frac{(1-p)w - \tilde{x}^\top \tilde{F}(\tilde{x})}{p} + k/p.$$

We refer the reader to [1] for a variety of practical conditions on the base game $\tilde{F}$ for which $\text{NE}(\tilde{F})$ is globally attracting under the replicator dynamics.

Corollary 2 shows that, not only does our control law succeed in eliminating cheating and consequently steering the population to a genuine, no cheating Nash equilibrium when the base dynamics naturally stabilize to them without the presence of cheating, but also, the penalty automatically assigned to cheating stabilizes to *almost* the smallest one necessary to make that no cheating state an equilibrium, i.e., $\frac{(1-p)w - \tilde{x}^\top \tilde{F}(\tilde{x})}{p}$. This can be thought of as our control law being $\frac{k}{p}$ -suboptimal in achieving our desired goal of eliminating cheating. Intuitively, this means that our control law is naturally “merciful” or “just;” it does not apply unnecessarily harsh penalties. This benevolent and economical feature of our control law is somewhat remarkable, as it was not designed *a priori* with this notion of optimality as a goal in mind. Certainly, the suboptimality gap k/p can be reduced by lowering the control gain k . However, this comes at the tradeoff of a slower rate of elimination of the adversarial strategy.

V. NUMERICAL SIMULATIONS

We now illustrate our model and results by simulating the augmented Rock-Paper-Scissors-Cheat game in Matlab, for both uncontrolled and controlled settings.

A. Uncontrolled Example

Consider the specific example given in Example 5, in which the RPSC game has three isolated Nash equilibria: $(1, 1, 1, 0)/3$, $(0, 0, 0, 1)$, and $(1, 1, 1, 4)/7$. We initialize the population state at $x(0) = (0.15, 0.05, 0.25, 0.55)$, so that, initially, 55% of the population is cheating. Simulating the replicator dynamics without any control (c is set to the constant value of 2) gives the population state trajectory shown by the thick gray curve in Figure 2. We observe for this initial condition that the population converges to the pure cheating equilibrium, e_4 (the red dot). Further simulation with different initial conditions reveals that no cheating, $(1, 1, 1, 0)/3$ (the blue dot), and pure cheating e_4 , both exhibit local asymptotic stability, whereas the relative interior Nash equilibrium $(1, 1, 1, 4)/7$ (on the orange line segment between pure cheating and no cheating) is unstable.

B. Controlled Examples

Using the same parameters and initial condition as in the uncontrolled simulation of Section V-A, we now simulate the closed-loop dynamics with the cheating penalty determined by our proposed control law (9) and (10). We use a control gain of $k = 1$, giving the trajectory shown in Figure 3. It is seen that, even though the initial population state naturally converges towards pure

Population Dynamics of RPSC

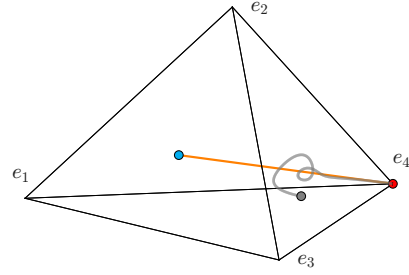


Fig. 2: The uncontrolled population dynamics converge to pure cheating.

cheating without control, the incorporation of our control law successfully steers the population to the no cheating Nash equilibrium $(1, 1, 1, 0)/3$. As guaranteed by Theorem 5, once cheating has been roughly eliminated (i.e., the population evolves along the e_1 - e_2 - e_3 face), the dynamics reduce back down to those of the base game (an asymptotically stable spiral in this case, since $w > l$; see [1, Example 6.3.1, Figure 6.7]). Re-simulating the closed-loop dynamics with various control gains k , we see in Figure 4 that the more aggressive controllers (larger k) result in fast dynamics (rapid elimination of cheating), at the expense of a suboptimal steady-state cheating penalty with suboptimality gap k/p , as predicted by Corollary 2.

Population Dynamics of RPSC

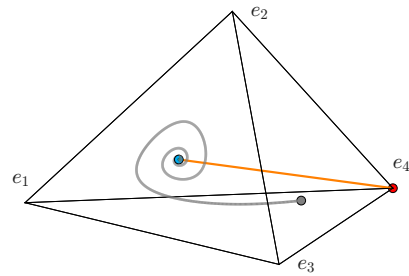


Fig. 3: The partial state feedback controller eliminates cheating, resulting in the controlled population dynamics converging to the unique no cheating Nash equilibrium.

We now consider modifying our parameters to $w = 1$, $l = 11$, $c = 10$, $p = 1/2$, $d_w = 0$, and $d_c = 9$, in particular so that $w < l$. In this case, the base game’s replicator dynamics approach a limit cycle tracing out the boundary of the e_1 - e_2 - e_3 face (see [1, Example 6.3.1, Figure 6.7]). A straightforward calculation shows that there are still three isolated Nash equilibria: $(1, 1, 1, 0)/3$, e_4 , and $(1, 1, 1, 7)/10$. We set the initial condition to $x(0) = (0.05, 0.10, 0.05, 0.80)$ and simulate the closed-loop dynamics with our control

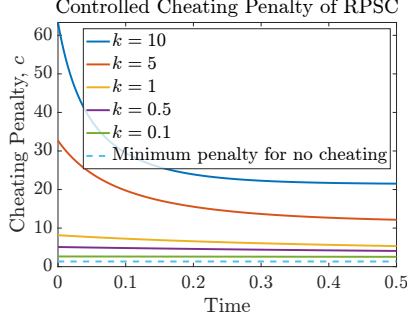


Fig. 4: Cheating penalty $c(t)$, as computed using our control law for various control gains k . The minimum penalty for no cheating is the value c_1 in Example 3.

law using $k = 1$. The resulting trajectory is shown in Figure 5. We observe that, as guaranteed by Theorem 5, our control law succeeds in eliminating cheating, and the population dynamics reduce down to genuine dynamics of the base game (i.e., the limit cycle).

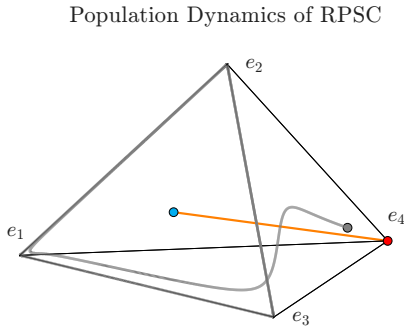


Fig. 5: The partial state feedback controller eliminates cheating, and the population dynamics reduce down to those of the base game (a limit cycle around Rock, Paper, and Scissors since $w < l$).

VI. CONCLUSIONS

This work proposes a new framework for modeling, analyzing, and controlling adversarial behaviors (such as cheating) in population games with linear payoff structures. We theoretically characterize all possible Nash equilibria of the game with cheating, and propose a partial feedback linearization control policy for adaptively setting the penalty assigned to cheating as a means to mitigate it. For populations evolving according to the classical replicator dynamics, we prove, and show through experimentation, that the proposed controller asymptotically eliminates cheating and recovers the dynamics of genuine gameplay according to the base game's rules, by judiciously resorting to penalties near the minimum required to achieve a no cheating equilibrium. A handful of interesting directions for future research remain, e.g., using the surveillance level

(the probability p) as the fundamental control variable, characterizing the social optimality and inefficiency of the various types of Nash equilibria of the augmented game, and extending the work to evolutionary dynamics other than the replicator dynamics.

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APPENDIX I
PROOFS

Theorem 1. *If $x \in \text{NE}(F)$, then x must lie on the line segment between the pure cheating population state e_{n+1} and some Nash equilibrium of the base game $\tilde{F}$:*

$$x = (1 - \theta)(\tilde{x}, 0) + \theta e_{n+1}, \quad (3)$$

for some $\theta \in [0, 1]$ and some $\tilde{x} \in \text{NE}(\tilde{F})$.

Proof of Theorem 1. Let $x \in \text{NE}(F)$. If $x_{n+1} = 1$, then $x = e_{n+1}$ and hence the result trivially holds true with $\tilde{x}$ being any arbitrary Nash equilibrium of $\tilde{F}$. Therefore, assume from here on that $x_{n+1} < 1$. Define $\theta = x_{n+1} \in [0, 1]$ and $\tilde{x} = \frac{1}{1-\theta}(x_1, \dots, x_n) \in \mathbb{R}^n$, so that

$$x = (x_1, \dots, x_n, 0) + x_{n+1}e_{n+1} = (1 - \theta)(\tilde{x}, 0) + \theta e_{n+1}.$$

It remains to prove that $\tilde{x} \in \text{NE}(\tilde{F})$.

Note that $\tilde{x} \geq 0$ and

$$\sum_{i=1}^n \tilde{x}_i = \sum_{i=1}^n \frac{1}{1-\theta} x_i = \frac{1}{1-\theta} \left(\sum_{i=1}^{n+1} x_i - x_{n+1} \right) = 1,$$

and therefore $\tilde{x} \in \Delta^{n-1}$. Since $x \in \text{NE}(F)$, it holds for all $i \in \{1, \dots, n\}$ with $x_i > 0$ that

$$(1 - \theta)\tilde{F}_i(\tilde{x}) + \theta(pw - (1 - p)l) = F_i(x) \geq F_j(x) = (1 - \theta)\tilde{F}_j(\tilde{x}) + \theta(pw - (1 - p)l)$$

for all $j \in \{1, \dots, n\}$. Therefore, it must be, for all $i \in \{1, \dots, n\}$ with $\tilde{x}_i = \frac{1}{1-\theta}x_i > 0$, that $\tilde{F}_i(\tilde{x}) \geq \tilde{F}_j(\tilde{x})$ for all $j \in \{1, \dots, n\}$. This proves that $\tilde{x} \in \text{NE}(\tilde{F})$. $\square$

Theorem 2. *Consider the pure cheating population state $x = e_{n+1}$. Assume that $p \in (0, 1)$. It holds that x is a Nash equilibrium of the game F if and only if*

$$c \leq \frac{(1 - p)l + (1 - p)^2 d_w - (w + d_c)p^2}{p(1 - p)}. \quad (4)$$

Proof of Theorem 2. Let $x = e_{n+1}$. It holds that

$$F(x) = \begin{bmatrix} (pw - (1 - p)l)\mathbf{1}_n \\ -p(pd_c + (1 - p)c) + (1 - p)(pw + (1 - p)d_w) \end{bmatrix}.$$

Therefore, $x = e_{n+1}$ is a Nash equilibrium if and only if

$$pw - (1 - p)l \leq -p(pd_c + (1 - p)c) + (1 - p)(pw + (1 - p)d_w)$$

which holds if and only if

$$p(1 - p)c + ((w + d_c)p^2 - (1 - p)l - (1 - p)^2 d_w) \leq 0. \quad (12)$$

Since $p \in (0, 1)$ by assumption, dividing by $p(1 - p)$ shows that this condition is equivalent to (4). $\square$

Theorem 3. *Let $x = (\tilde{x}, 0)$ be a population state without cheating, and with $\tilde{x} \in \text{NE}(\tilde{F})$. It holds that x is a Nash equilibrium of F if and only if*

$$-pc + (1 - p)w \leq \tilde{x}^\top \tilde{F}(\tilde{x}). \quad (5)$$

Proof of Theorem 3. Suppose first that $x \in \text{NE}(F)$. Then it holds that

$$y^\top F(x) \leq x^\top F(x) \text{ for all } y \in \Delta^n.$$

Choosing $y = e_{n+1}$, this implies that

$$F_{n+1}(x) \leq x^\top F(x).$$

Since $x = (\tilde{x}, 0)$, this implies that

$$(-pc + (1 - p)w)\mathbf{1}_n^\top \tilde{x} \leq \tilde{x}^\top \tilde{F}(\tilde{x}),$$

which proves (5) since $\mathbf{1}_n^\top \tilde{x} = 1$.

On the other hand, suppose that (5) holds. Then

$$-pc + (1-p)w \leq \tilde{x}^\top \tilde{F}(\tilde{x}) + 0 \cdot (-pc + (1-p)w) = x^\top F(x)$$

since $x = (\tilde{x}, 0)$. Therefore,

$$\begin{aligned} x^\top F(x) &\geq \mathbf{0}_n^\top \tilde{F}(\tilde{x}) + 1 \cdot (-pc + (1-p)w) \\ &= e_{n+1}^\top F(x), \end{aligned} \quad (13)$$

where $\mathbf{0}_n$ is the n -vector of all zeros. Furthermore, since $\tilde{x} \in \text{NE}(\tilde{F})$, it holds that

$$\begin{aligned} x^\top F(x) &= \tilde{x}^\top \tilde{F}(\tilde{x}) + 0 \cdot (-pc + (1-p)w) \\ &\geq \tilde{y}^\top \tilde{F}(\tilde{y}) + 0 \cdot (-pc + (1-p)w) \\ &= (\tilde{y}, 0)^\top F(x) \end{aligned}$$

for all $\tilde{y} \in \Delta^{n-1}$. Combining the two previous inequalities, we find that

$$\begin{aligned} x^\top F(x) &= (1-\theta)x^\top F(x) + \theta x^\top F(x) \\ &\geq (1-\theta)(\tilde{y}, 0)^\top F(x) + \theta e_{n+1}^\top F(x) \\ &= ((1-\theta)\tilde{y}, \theta)^\top F(x) \end{aligned} \quad (14)$$

for all $\theta \in [0, 1]$ and all $\tilde{y} \in \Delta^{n-1}$.

Now, let $y \in \Delta^n$ be arbitrary. If $y_{n+1} = 1$, then $y = e_{n+1}$ and hence (13) implies that $x^\top F(x) \geq y^\top F(x)$. On the other hand, suppose that $y_{n+1} < 1$. Choose $\theta = y_{n+1} \in [0, 1]$ and $\tilde{y} = (y_1, \dots, y_n)/(1-\theta)$. Then, (14) implies that

$$x^\top F(x) \geq \left((1-\theta) \frac{(y_1, \dots, y_n)}{1-\theta}, \theta \right)^\top F(x) = y^\top F(x).$$

Therefore, $x \in \text{NE}(F)$. □

Theorem 4. *If x is a Nash equilibrium of F in the relative interior of Δ^n , then it must take the form (3) for some $\tilde{x} \in \text{NE}(\tilde{F})$, with $\theta \in (0, 1)$ satisfying the affine equation*

$$\mathcal{A}\theta = b. \quad (7)$$

where

$$\begin{aligned} \mathcal{A} &:= p^2 c + \left(\frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x} - (1-p+p^2)w \right. \\ &\quad \left. + (1-p)l + (1-p)^2 d_w - p^2 d_c \right), \\ b &:= pc + \left(\frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x} - (1-p)w \right). \end{aligned}$$

In this case, either the relative interior Nash equilibrium is unique with

$$\theta = \mathcal{A}/b \in (0, 1), \quad (8)$$

or every point on the line segment $\{(1-\theta)(\tilde{x}, 0) + \theta e_{n+1} : \theta \in (0, 1)\}$ is a Nash equilibrium of F .

On the other hand, if $\tilde{x} \in \text{NE}(\tilde{F})$ and if $\theta \in (0, 1)$ satisfies (7), then x given by (3) is a relative interior Nash equilibrium of F .

Proof of Theorem 4. Let $x \in \Delta^n$ be a relative interior Nash equilibrium of the augmented game F . Then, it holds that $F_i(x) = \max_{j \in \{1, \dots, n+1\}} F_j(x)$ for all $i \in \{1, \dots, n+1\}$. By Theorem 1, it must be the case that x takes the form (3), i.e.,

$$x = (1-\theta)(\tilde{x}, 0) + \theta e_{n+1},$$

for some $\theta \in (0, 1)$ and some $\tilde{x} \in \text{NE}(\tilde{F})$. It remains to determine θ .

Since $\tilde{x} \in \text{NE}(\tilde{F})$ and $\tilde{x}_i > 0$ for all $i \in \{1, \dots, n\}$, it holds for all $i \in \{1, \dots, n\}$ that $\tilde{a}_i^\top \tilde{x} = \frac{1}{n} \sum_{j=1}^n \tilde{a}_j^\top \tilde{x} = \frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x}$, and therefore that

$$\begin{aligned} F_i(x) &= (1-\theta) \tilde{a}_i^\top \tilde{x} + \theta (pw - (1-p)l) \\ &= (1-\theta) \cdot \frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x} + \theta (pw - (1-p)l). \end{aligned}$$

This must equal

$$F_{n+1}(x) = (1 - \theta)(-pc + (1 - p)w)\mathbf{1}_n^\top \tilde{x} + \theta(-p(pd_c + (1 - p)c) + (1 - p)(pw + (1 - p)d_w)). \quad (15)$$

Setting $F_i(x) = F_{n+1}(x)$ using the above expressions, we arrive at the affine equation (7) in the variable θ . Either θ is the unique solution to this equation, or the equation is solved by every real number. In the former case, it is clear that x is the only relative interior Nash equilibrium. To prove that every point on the line segment between $(\tilde{x}, 0)$ and e_{n+1} is a Nash equilibrium in the latter case, we first prove the reverse implication (i.e., the final assertion of Theorem 4).

To prove the reverse implication, let $\tilde{x} \in \text{NE}(\tilde{F})$ and suppose that $\theta \in (0, 1)$ satisfies (7). We will show that $x = (1 - \theta)(\tilde{x}, 0) + \theta e_{n+1}$ is a relative interior Nash equilibrium of F . First, it is clear that x is in the relative interior of Δ^n . Next, we note that θ solves the equation

$$F_{n+1}(x) = (1 - \theta) \cdot \frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x} + \theta(pw - (1 - p)l)$$

with $F_{n+1}(x)$ given in terms of θ by (15). Finally, we note that, since $\tilde{x} \in \text{NE}(\tilde{F})$, it holds that $\tilde{a}_i^\top \tilde{x} = \tilde{a}_j^\top \tilde{x}$ for all $i, j \in \{1, \dots, n\}$, implying that $\tilde{a}_i^\top \tilde{x} = \frac{1}{n} \sum_{j=1}^n \tilde{a}_j^\top \tilde{x} = \frac{1}{n} \mathbf{1}_n^\top \tilde{A} \tilde{x}$ for all $i \in \{1, \dots, n\}$, further implying for all such i that

$$F_i(x) = (1 - \theta)\tilde{a}_i^\top \tilde{x} + \theta(pw - (1 - p)l) = F_{n+1}(x).$$

Therefore, we conclude that $F_i(x) = \max_{j \in \{1, \dots, n+1\}} F_j(x)$ for all $i \in \{1, \dots, n+1\}$, and hence x is a relative interior Nash equilibrium. This immediately proves the deferred assertion that every point on the line segment between $(\tilde{x}, 0)$ and e_{n+1} is a Nash equilibrium of F if the affine equation (7) admits every real number as a solution. $\square$

Corollary 1. Consider a Nash equilibrium $\tilde{x} \in \text{NE}(\tilde{F})$ of the base game $\tilde{F}$. If θ defined by (8) is a well-defined real number and $\theta \notin (0, 1)$, then there are no Nash equilibria of the augmented game F belonging to the strict line segment between $(\tilde{x}, 0)$ and e_{n+1} .

Proof of Corollary 1. This follows immediately from Theorem 4. $\square$

Theorem 5. Let $\tilde{f}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the vector field defining the replicator dynamics of the base game, and let $f: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ be the vector field defining the closed-loop replicator dynamics of the augmented game with the control law (9) and (10) with $k > 0$:

$$\begin{aligned} \tilde{f}_i(\tilde{x}) &= \tilde{x}_i \left(\tilde{F}_i(\tilde{x}) - \tilde{x}^\top \tilde{F}(\tilde{x}) \right), \\ f_j(x) &= x_j (g_j^\top x - x^\top Gx) \\ &\quad - x_j (h_j^\top x - x^\top Hx) \frac{g_{n+1}^\top x - x^\top Gx + k}{h_{n+1}^\top x - x^\top Hx}, \end{aligned}$$

for $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, n+1\}$. Define the shorthand notation $x_{1:n} = (x_1, \dots, x_n)$. If $p > 0$, then the closed-loop trajectory of the augmented replicator dynamics satisfies both of the following:

- 1) $\lim_{t \rightarrow \infty} |c(t) - \tilde{c}(t)| = 0$, where

$$\tilde{c}(t) := \frac{(1 - p)w - x_{1:n}(t)^\top \tilde{A} x_{1:n}(t) + k}{p}, \quad (11)$$

and

- 2) $\lim_{t \rightarrow \infty} |f_i(x(t)) - \tilde{f}_i(x_1(t), \dots, x_n(t))| = 0$ for all $i \in \{1, \dots, n\}$.

Proof of Theorem 5. To streamline presentation throughout the proof, we do not explicitly denote dependence on time, and we use the shorthand notation $x_{1:n} = (x_1, \dots, x_n)$ and $\gamma = A_{n+1, n+1} = -p^2 d_c + (1 - p)(pw + (1 - p)d_w)$. Notice the following:

$$\begin{aligned} g_{n+1}^\top x &= (1 - p)w \mathbf{1}_n^\top x_{1:n} + \gamma x_{n+1}, \\ x^\top Gx &= x_{1:n}^\top \tilde{A} x_{1:n} + (pw - (1 - p)l) x_{n+1} \mathbf{1}_n^\top x_{1:n} \\ &\quad + (1 - p)w x_{n+1} \mathbf{1}_n^\top x_{1:n} + \gamma x_{n+1}^2, \\ h_{n+1}^\top x &= -p \mathbf{1}_n^\top x_{1:n} - p(1 - p)x_{n+1}, \\ x^\top Hx &= -p x_{n+1} \mathbf{1}_n^\top x_{1:n} - p(1 - p)x_{n+1}^2. \end{aligned} \quad (16)$$

Therefore, since $h_{n+1}^\top x \xrightarrow{t \rightarrow \infty} -p$ and $x^\top Hx \xrightarrow{t \rightarrow \infty} 0$, we find that

$$\begin{aligned} \lim_{t \rightarrow \infty} |c - \tilde{c}| &= \lim_{t \rightarrow \infty} \left| \frac{-g_{n+1}^\top x + x^\top Gx - k}{h_{n+1}^\top x - x^\top Hx} - \frac{(1-p)w - x_{1:n}^\top \tilde{A}x_{1:n} + k}{p} \right| \\ &= \lim_{t \rightarrow \infty} \left| \frac{1}{p(h_{n+1}^\top x - x^\top Hx)} \left(-pg_{n+1}^\top x + px^\top Gx - pk \right. \right. \\ &\quad \left. \left. - (h_{n+1}^\top x - x^\top Hx)((1-p)w - x_{1:n}^\top \tilde{A}x_{1:n} + k) \right) \right| \\ &= \frac{1}{p^2} \lim_{t \rightarrow \infty} \left| -pg_{n+1}^\top x + px^\top Gx - pk \right. \\ &\quad \left. - (h_{n+1}^\top x - x^\top Hx)((1-p)w - x_{1:n}^\top \tilde{A}x_{1:n} + k) \right|. \end{aligned}$$

Substituting in the expressions from (16) and applying $x_{n+1} \xrightarrow{t \rightarrow \infty} 0$ and $\mathbf{1}_n^\top x_{1:n} \xrightarrow{t \rightarrow \infty} 1$, one finds that every term in the limit either approaches zero or is bounded and exactly drops out of the expression, leading to $|c - \tilde{c}| \xrightarrow{t \rightarrow \infty} 0$, which proves the first assertion. To prove the second assertion, let $i \in \{1, \dots, n\}$. Notice the following:

$$\begin{aligned} g_i^\top x &= \tilde{a}_i^\top x_{1:n} + (pw - (1-p)l)x_{n+1}, \\ h_i^\top x &= 0. \end{aligned}$$

Therefore, since $|x_i| \leq 1$, and since $x_{n+1} \xrightarrow{t \rightarrow \infty} 0$ and hence $\mathbf{1}_n^\top x_{1:n} \xrightarrow{t \rightarrow \infty} 1$, we find that

$$\begin{aligned} \lim_{t \rightarrow \infty} |f_i(x) - \tilde{f}_i(x_{1:n})| &= \lim_{t \rightarrow \infty} |x_i(g_i^\top x - x^\top Gx) \\ &\quad - x_i(h_i^\top x - x^\top Hx)c \\ &\quad - x_i(\tilde{a}_i^\top x_{1:n} - x_{1:n}^\top \tilde{A}x_{1:n})| \\ &= \lim_{t \rightarrow \infty} |x_i(pw - (1-p)l)x_{n+1} \\ &\quad - x_i(pw - (1-p)l)x_{n+1}\mathbf{1}_n^\top x_{1:n} \\ &\quad - x_i(1-p)wx_{n+1}\mathbf{1}_n^\top x_{1:n} - x_i\gamma x_{n+1}^2 \\ &\quad - x_i(h_i^\top x - x^\top Hx)c| \\ &= \lim_{t \rightarrow \infty} |x_i(h_i^\top x - x^\top Hx)c| \\ &\leq \lim_{t \rightarrow \infty} |x^\top Hx|(|c - \tilde{c}| + |\tilde{c}|) \\ &\leq \lim_{t \rightarrow \infty} |x^\top Hx| \frac{(1-p)w + |x_{1:n}^\top \tilde{A}x_{1:n}| + k}{p} \\ &= 0, \end{aligned}$$

where the second-to-last inequality uses the fact that $|c - \tilde{c}| \xrightarrow{t \rightarrow \infty} 0$ and the final equality follows from the boundedness of $x_{1:n}^\top \tilde{A}x_{1:n}$ together with the fact that $x^\top Hx \xrightarrow{t \rightarrow \infty} 0$. This concludes the proof. $\square$

Corollary 2. *Suppose that $\text{NE}(\tilde{F})$ is globally attracting under the base game's replicator dynamics. Then, the closed-loop replicator dynamics of the augmented game with the control law (9) and (10) converge to a lifted Nash equilibrium $(\tilde{x}, 0)$, $\tilde{x} \in \text{NE}(F)$, with no cheating:*

$$\lim_{t \rightarrow \infty} x(t) = (\tilde{x}, 0).$$

Furthermore, the time-varying penalty assigned to cheating converges to the minimum value necessary to ensure the no cheating equilibrium, determined by (5) in Theorem 3, offset by a constant value of k/p :

$$\lim_{t \rightarrow \infty} c(t) = \frac{(1-p)w - \tilde{x}^\top \tilde{F}(\tilde{x})}{p} + k/p.$$

Proof of Corollary 2. The fact that $x(t) \xrightarrow{t \rightarrow \infty} (\tilde{x}, 0)$ for some $\tilde{x} \in \text{NE}(\tilde{F})$ follows immediately from Theorem 5, since $x_{n+1}(t) \xrightarrow{t \rightarrow \infty} 0$ and the dynamics f_i governing the evolution of the remaining population states x_i , $i \in \{1, \dots, n\}$, converge to the dynamics $\tilde{f}$ of the uncontrolled base game which steer $(x_1, \dots, x_n)$ to a Nash equilibrium of $\tilde{F}$. The convergence of the cheating penalty also follows immediately by remarking that $|c(t) - \tilde{c}(t)| \xrightarrow{t \rightarrow \infty} 0$ and $\tilde{c}(t) \xrightarrow{t \rightarrow \infty} \frac{(1-p)w - \tilde{x}^\top \tilde{A}\tilde{x} + k}{p} = \frac{(1-p)w - \tilde{x}^\top \tilde{F}(\tilde{x})}{p} + k/p$ per (11). $\square$